



DATA INFRASTRUCTURE & COMPUTE PROFILE

Ukrainian Legal Intelligence Platform

legal.org.ua · SecondLayer · ТОВ "Лекс ЕйАй"

legal.org.ua

132M

COURT DECISIONS

55M

REGISTRY RECORDS

570B

EMBEDDING TOKENS

12.3TB

VECTOR STORE

1

Data Infrastructure

Overview

3 databases · ~152 GB active · 18.8 TB target

132M

COURT DECISIONS
NATIONAL REGISTRY

8.87M

BUSINESS ENTITIES
ЄДР + ФОП

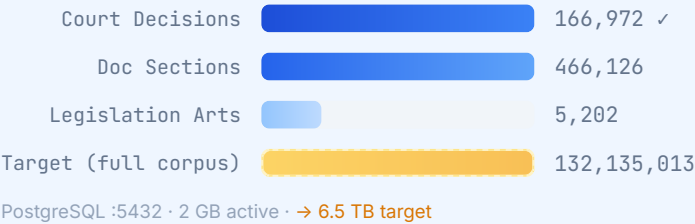
30.6M

ENFORCEMENT
PROCEEDINGS

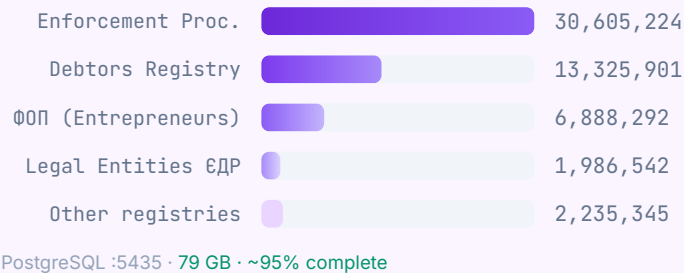
50K

LAWS & CODES
LEGISLATION

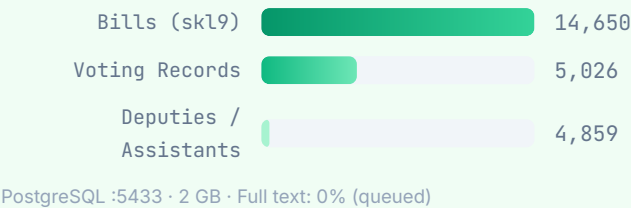
🏛 MCP_BACKEND — COURT & LEGISLATION



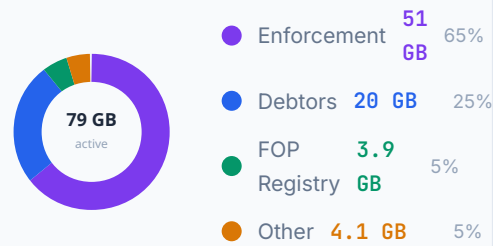
🏛 MCP_OPENREYESTR — STATE REGISTRIES



🏛 MCP_RADA — PARLIAMENT DATA



STORAGE DISTRIBUTION (ACTIVE 79 GB)



UPDATE FREQUENCIES

SOURCE	FREQ
Court Decisions	ON-DEMAND
ΕΔΡ / ΦΟΠ	DAILY XML
Enforcement	DAILY CSV
Debtors	DAILY CSV
Bankruptcy	DAILY XML
RADA Bills	DAILY API
RADA Voting	DAILY API
Notaries/Experts	WEEKLY XML
Legislation	MANUAL SYNC

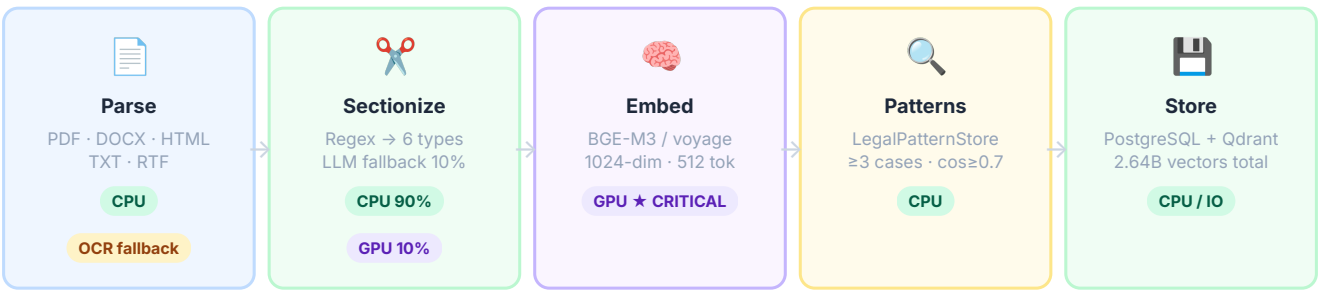
COURT DECISION PROFILE (REAL DATA)

Avg length	21,073 chars
Avg tokens	5,268 tokens
Sections / doc	19 avg
Section size	485 chars
Dominant type	LAW_REFS (53%)
Import speed	19,744 docs/day

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Document Processing Pipeline

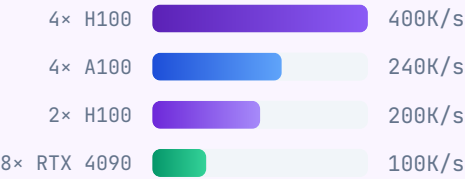
5-stage · parse → section → embed → pattern → store



HARDWARE COMPARISON — BULK PROCESSING TIMELINE

1× H100 80GB 66 days embedding 720 days LLM bottleneck	4× H100 (embed) + API LLM 17 days embedding ✓ \$30K LLM via API ★ RECOMMENDED	4× H100 (embed) + 4× H100 (LLM) 17 days embedding ✓ 35 days LLM parallel ✓	4× A100 80GB + API LLM 27 days embedding ✓ \$30K LLM via API	8× RTX 4090 + API LLM 66 days embedding ⚠ \$30K LLM via API
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EMBEDDING THROUGHPUT (TOK/SEC · BATCH MODE)



💰 BULK PROCESSING COST (ONE-TIME)

All via Cloud API **\$30,087**

VoyageAI + OpenAI gpt-4o-mini

Embed API + Self-host LLM **\$13,415**

VoyageAI + Qwen 72B on 4× H100 (35 days elec.)

★ Cloud rental 8× H100 · 35 days **\$7,200**

8 GPUs × 35d × 24h × \$2.50/h → then buy 4× H100

Self-hosted 8× H100 (capex) **\$200K**

\$200K hardware, 5+ years, ~\$7K electricity/year

ONGOING COMPUTE (1K USERS DAILY)

WORKLOAD	VOLUME/DAY
Real-time LLM inference	~75M tokens
New court decisions	5K docs → embed
OpenReyestr delta	~50K records
RADA updates	~200 records
GPU Required	4× H100 sufficient

STORAGE: ACTIVE → FULL DATASET

COMPONENT	NOW	TARGET
PostgreSQL (Backend)	2 GB	6.5 TB
PostgreSQL (OpenReyestr)	79 GB	~160 GB
PostgreSQL (RADA)	2 GB	~45 GB
Qdrant (Court)	<1 GB	10.8 TB
Qdrant (Other)	<1 GB	1.5 TB
TOTAL	~83 GB	~18.8 TB

MODEL SELECTION

TASK	MODEL
Inference (chat/reasoning)	QWEN 2.5 72B
Inference (fallback)	LLAMA 3.3 70B
Bulk embeddings	BGE-M3 (1024-DIM)
Doc parsing / OCR	GOOGLE VISION API

TOGETHER.AI USAGE PROJECTION

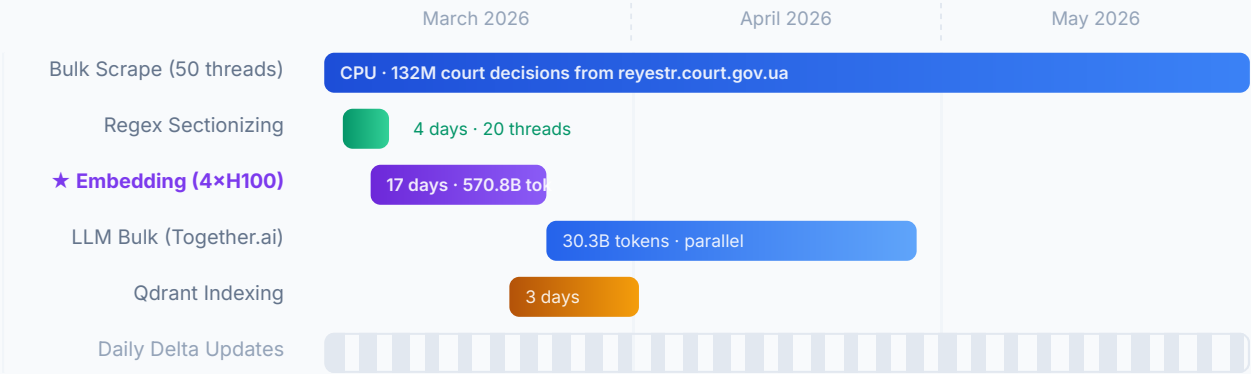
PERIOD	REQ/DAY	TOKENS/DAY
Now (3 users)	50-100	250K
Bulk processing (3 mo)	batch	~337M
Post-launch (1K users)	15-20K	75M/day

Primary: Qwen 2.5 72B Instruct · Llama 3.3 70B
30.3B token bulk batch over first 3 months

3-Month Bulk Processing Roadmap

Target: 132M court decisions fully indexed · May 2026

EXECUTION GANTT — MARCH · APRIL · MAY 2026



Month 1 — March 2026: Data Acquisition

- Launch 50-thread bulk scraper for reyestr.court.gov.ua
- Target: 1M court decisions/day via ZakonOnline API
- Regex sectionizing pipeline: 20 threads → 4 days
- OpenReyestr: complete debtors (+3M) & enforcement (+1.8M) delta
- Start embedding on first batch of 20M completed docs

Month 2 — April 2026: Embedding Blitz

- 4× H100 dedicated to BGE-M3 @ 400K tokens/sec
- Complete all 570.8B tokens in 17 days
- Qdrant: index 2.64B vectors → 10.8 TB vector store
- Together.ai API: 30.3B token LLM batch (sectionizer fallback)
- RADA: load bill full texts skl6–skl9 (14,650 bills)

Month 3 — May 2026: Enrichment & Scale

- LegalPatternStore: extract patterns across 132M decisions
- Beneficiary chains: 773K entity analysis via LLM
- Legislation: embed 50K laws (947M tokens, <1 day)
- QA validation: semantic search accuracy, citation testing
- Launch: scale from 3 → 50 → 1,000 attorney users

3-MONTH SUMMARY: BY THE NUMBERS

METRIC	TARGET
Court decisions indexed	132,135,013
Semantic sections stored	2,640,000,000
Qdrant vectors total	2,650,000,000
Embedding tokens (bulk)	570.8B
LLM tokens (API batch)	30.3B
PostgreSQL total storage	~6.7 TB
Qdrant vector storage	~12.3 TB
Total storage footprint	~18.8 TB

PLATFORM STATUS

ITEM	STATUS
Production deployment	LIVE
Active attorney users	3 (EARLY ACCESS)
OpenReyestr completeness	~95%
Court decisions indexed	0.13% OF FULL
Legislation corpus	12 CODES LOADED
RADA bills full text	0% (QUEUED)

COMPANY

TOB "Лекс ЕйАй" · Ukraine
Domain: legal.org.ua
Billing admin: vovkes@legal.org.ua
Primary language: Ukrainian / Russian
Stack: Node.js · PostgreSQL · Qdrant · Redis
LLM: OpenAI → migrating to Together.ai